Department of Statistics And Data Science
CHRIST (DEEMED TO BE UNIVERSITY)

Continuous Internal Assessment 3

Waste Image Classification Using Machine Learning with HOG, LBP, and GLCM Features

Subject:
DSC301-4 Machine Learning

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Acknowledgements

First of all, we would like to thank God for giving us the strength and energy to work on this project with full rigor.

We would like to extend a big thank you to my Professor of Machine Learning, Dr. Umme Salma M, for her thorough guidance and encouragement in the creation of this assignment.

We would like to thank our peers and family for the constant support and guidance which enabled us to complete this project on time. Further, we would like to thank the Department of Statistics and Data Science for giving us the best environment to study and for integrating industry relevant projects of this kind. We hope we were able to uphold the university's vision of Excellence and Service.

1 Overview

This project investigates the classification of household waste images using classical Machine Learning Approaches. By extracting meaning color, texture and shape features - such as Histogram of Oriented Gradients(HOG), Local Binary Patterns(LBP) and Gray Level Co-occurance Matrix (GLCM) - the study aims to enhance the discriminative capabilities of traditional classifiers such as Logistics Regression, Decision Tree and Random Forest. The findings emphasize the critical role of effective feature engineering in improving performance in image based classification tasks without relying on Deep Learning models.

2 Introduction

Waste management is a significant challenge in the modern era due to the increasing volume of waste generated on a daily basis. Proper waste segregation plays a crucial role in promoting recycling and minimizing environmental pollution. However, traditional manual waste segregation methods are time-consuming, costly, and prone to human error. To address these limitations, automated waste classification using machine learning techniques has gained increasing attention [1].

In image-based waste classification, images of waste items are analyzed to identify their respective categories. For accurate classification, machine learning models require informative features extracted from images. Feature extraction helps capture important visual characteristics such as color, shape, and texture, which are essential for distinguishing between different types of waste [1].

In this project, Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and Gray Level Co-occurrence Matrix (GLCM) features are employed. These feature extraction techniques are widely used in image processing as they effectively represent structural and texture information present in images [2, 3, 4]. The extracted features are then used to train classical machine learning models for binary waste classification.

3 Background & Problem Statement

3.1 Background

Waste management has emerged as a major social and environmental challenge due to the rapid increase in the generation of solid waste, especially in urban areas. Improper waste segregation reduces recycling efficiency and contributes to environmental pollution, making sustainable waste handling a critical concern. These challenges are directly related to the objectives of **Sustainable Development Goals (SDG) 11- Sustainable Cities and Communities** and **SDG12- Responsible Consumption and Production**, which emphasize efficient waste management and resource utilization. Traditional waste segregation methods rely heavily on manual sorting, which is time-consuming, costly, and prone to human error. To address these limitations, automated waste classification systems using machine learning techniques have gained increasing attention [1].

Image-based waste classification involves analyzing visual information from images of waste items to determine their category. Such automated systems can support improved segregation at the source, thus improving recycling rates and reducing landfill usage, aligning with the sustainability objectives outlined in the SDGs. However, the effectiveness of image-based classification systems is largely dependent on the quality of features extracted from the images. Feature extraction plays a crucial role in the transformation of raw image data into meaningful numerical representations that can be processed by machine learning models.

Classical feature extraction techniques remain relevant due to their interpretability and computational efficiency, making them suitable for practical waste management applications. Histogram of Oriented Gradients (HOG) captures structural and shape-related information by analyzing gradient orientations in images [2]. Local Binary Patterns (LBP) are effective in representing texture through local intensity variations [3], while Gray Level Co-occurrence Matrix (GLCM) features provide statistical measures of texture based on pixel intensity relationships [4]. These features collectively enable machine learning models to distinguish between different waste types.

Recent datasets developed for waste classification further highlight the growing research interest in automated waste segregation systems [5]. By leveraging classical machine learning techniques and effective feature engineering, this project contributes to sustainable waste management practices and indirectly supports **SDG 13 – Climate Action** by promoting improved recycling and reduced environmental pollution.

3.2 Problem Statement

Manual waste segmentation is inefficient and unreliable, leading to improper disposal and reduced recycling effectiveness. There is a need for an automated system that can accurately classify waste items into recyclable and non recyclable categories using image based features and classical machine learning approach.

Assumptions:

1. This study is restricted to binary classification of waste into recyclable and non-recyclable categories.
2. The system operates on static images of waste items captured under reasonable lighting conditions.
3. The project focuses on classical machine learning models, excluding deep learning approaches.

3.3 Need

The primary need for doing this project is **Social**. The project aims to improve the current manual segmentation procedures carried by municipalities and tends to reduce the error in manual segmentation.
The project additionally also prioritizes a **conceptual** need with the aim of extraction of HOG, LBP and GLCM features on very few images and gain maximum class separability.

3.4 Relevance

This project is relevant to the domains of **environmental sustainability and smart waste management**. It demonstrates the practical application of classical machine learning techniques in solving real-world problems. Additionally, the project emphasizes the importance of feature engineering in image-based classification, making it suitable for academic learning as well as real-world implementation scenarios where computational resources may be limited.

4 Learning Objectives & Outcomes

4.1 Learning Objectives

The objectives of this project were to:

1. Identify relevant visual features for image-based waste classification
2. Apply data preprocessing techniques to prepare image-derived features
3. Implement classical machine learning classification algorithms
4. Analyze model performance using appropriate evaluation metrics
5. Interpret results to understand class separability and prediction behavior.

4.2 Outcomes

Upon successful completion of the project, the following outcomes were realized:

1. A structured data set that contains extracted features from waste images suitable for machine learning tasks.
2. An automated waste classification system capable of distinguishing between recyclable and non-recyclable waste.
3. Comparative performance analysis of multiple classification models based on accuracy, precision, recall, and F1-score.
4. Insights into the effectiveness of classical feature extraction techniques for image-based classification.
5. Practical understanding of how feature engineering and preprocessing influence model performance in real-world applications.

5 Dataset Source and Description

5.1 Data Source

We have made use of both primary and secondary data sources. The project makes use of both stratified and purposive sampling techniques to collect the images. To enhance the project, secondary data was also utilized. [5]

The secondary dataset has been collected with the following credentials:
Link: https://ieee-dataport.org/documents/multi-class-waste-classification-image-dataset-cnn-training-and-tflite-deployment-edge
Original Citation:
Aarav Rao, "Multi-Class Waste Classification Image Dataset for CNN Training and TFLite Deployment on Edge Devices", IEEE Dataport, June 30, 2025, doi:10.21227/mqyq-wd15
Cited by:
The dataset is fairy new, last updated on 11th of January, 2026. Thus, has not yet been cited by researchers. But, having been uploaded on IEEE Dataport, the dataset is of rich and will definitely be used in the future by researchers.

5.2 Dataset Description

Two classes have been created- Recyclable and Non-Recyclable. The Recyclable class has 510 image samples and the non-recyclable class has 501 image samples. Thus, a total of 1011 images have been sampled. The dataset is perfect for classification-based problem.

6 Methodology

This project follows the Knowledge Discovery in Databases (KDD) process to systematically extract meaningful insights from image data. The process begins with data collection, where waste images are gathered and organized into recyclable and non-recyclable categories. This is followed by data preprocessing, which includes feature extraction, exploratory data analysis, outlier handling, and feature scaling to prepare the data for modeling. Dimensionality reduction is then applied to improve class separability. Finally, machine learning models are trained, evaluated, and compared using standard performance metrics. This structured approach ensures that the transformation from raw image data to actionable knowledge is both systematic and reproducible.

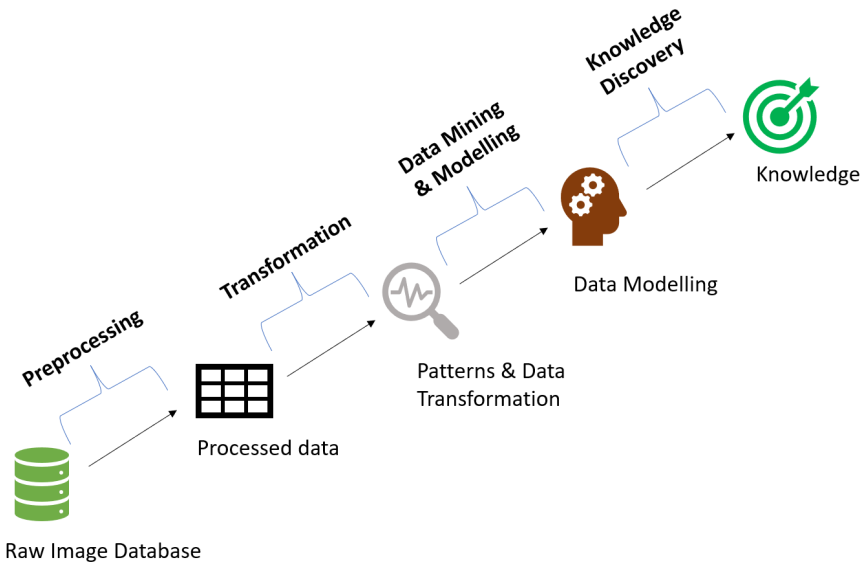

Figure 1: Knowledge Discovery in Databases (KDD) process

6.1 Data Collection

The dataset consists of images of household waste items collected primarily from reliable secondary sources, supplemented with a limited number of primary samples. The images were categorized into two classes: **recyclable and non-recyclable**. All images were organized into class-wise folders, where folder names were used as labels during feature extraction. This ensured consistency and ease of dataset handling.

6.2 Feature Extraction

Since machine learning models cannot directly process raw images, meaningful features were extracted to convert images into numerical form. The following feature extraction techniques were used:

- **Color Features:** Mean values of RGB (Red Green Blue) channels were extracted to capture color information.
- **Histogram of Oriented Gradients (HOG):** Used to extract shape and edge-related features. [2]
- **Local Binary Patterns (LBP):** Used to capture local texture patterns in images.[3]
- **Gray Level Co-occurrence Matrix (GLCM):** Texture features such as contrast, homogeneity, energy, and correlation were extracted.[4]

All extracted features were then combined into a structural tabular CSV file for further analysis.

6.3 Exploratory Data Analysis and Data Preprocessing

6.3.1 Exploratoray Data Analysis

Exploratory Data Analysis is a mechanism of understanding data, identifying patterns and behaviour of data either by using Statistical techniques or visual methods. EDA helps us to understand the data in a better way so that we choose the best model for implementation that gives maximum performance. As we can observe from fig 2, no

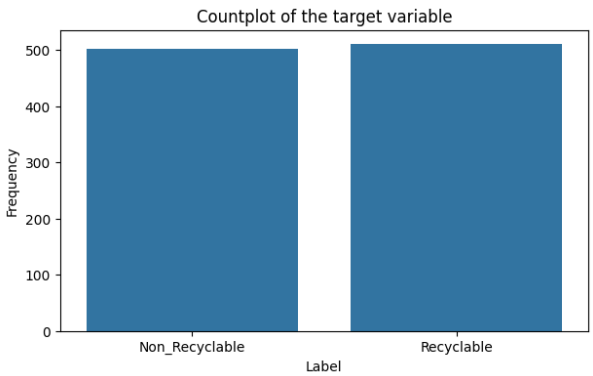

Figure 2: Countplot of the target variable

class imbalance exists. Both classes have a balanced number of images. The box plots

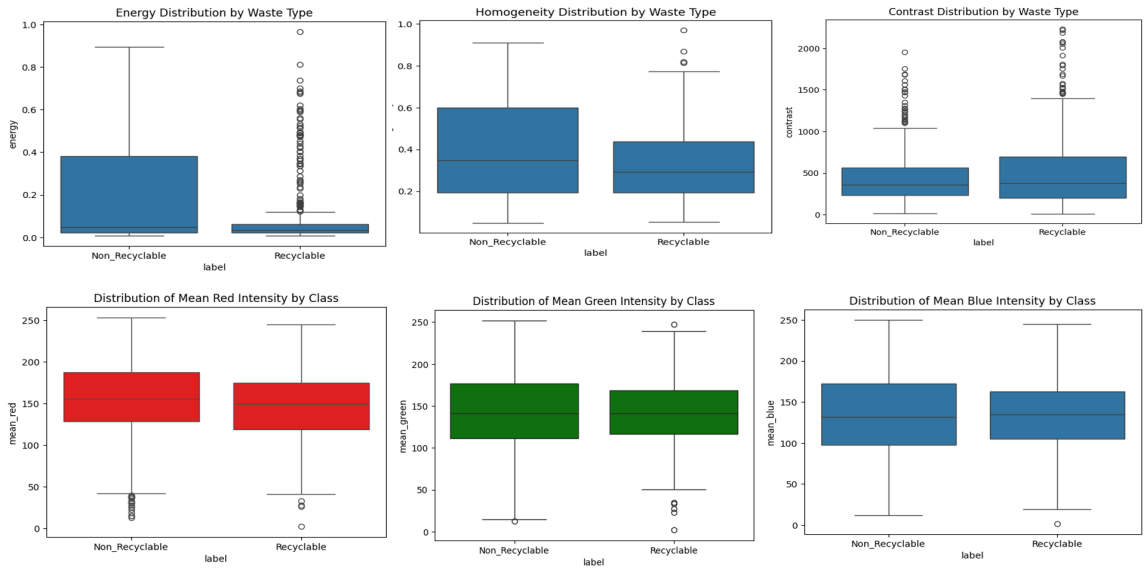

Figure 3: Boxplot Analysis

illustrate the distribution of certain color and texture features for recyclable and non-recyclable waste. Certain texture features like GLCM energy, homogeneity, and contrast demonstrate a clear difference between the two classes, thus establishing their effectiveness in identifying differences in waste materials based on texture. Recyclable waste tends to have higher contrast and variability, which is expected in materials like plastic, metal, and glass.

On the other hand, mean RGB intensity features demonstrate a considerable overlap between the two classes, thus establishing the fact that color features alone are not sufficient for efficient classification. However, these features can still be used as supplementary information when combined with texture features.

The EDA analysis suggests that texture features are more discriminative than color features, and a combination of multiple feature types is required for efficient classification. The existence of overlapping regions and outliers establishes the need for certain preprocessing steps like feature scaling and dimensionality reduction prior to training.

6.3.2 Data Preprocessing

Preprocessing steps were applied to improve model performance and reliability. Outliers in numerical features were handled using the Interquartile Range (IQR) capping method, particularly to reduce the impact of extreme values on Logistic Regression.

$$\text{IQR} = Q_3 - Q_1$$

$$\text{Lower Outlier Threshold} = Q_1 - (1.5 \times \text{IQR})$$

$$\text{Upper Outlier Threshold} = Q_3 + (1.5 \times \text{IQR})$$

Feature scaling was performed using Standardization to ensure a uniform contribution of features.

$$z = \frac{x - \mu}{\sigma}$$

Additionally, Linear Discriminant Analysis (LDA) was applied for dimensionality reduction and to enhance class separability.LDA is a supervised dimensionality reduction technique that maximizes class separability. Considering the existence of more than 3000 extracted features, we have opted for Linear Discriminant analysis (LDA). We obtain a single feature which maximizes class separability.

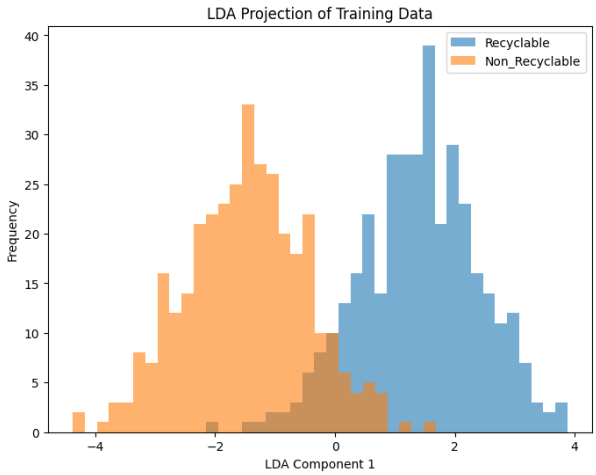

Figure 4: LDA Projection of Training Data

6.4 Data Mining

Before applying machine learning algorithms, the dataset was divided into training and testing subsets using the train–test split approach. This step is essential to evaluate how well the trained models generalize to unseen data. In this project, the dataset was split such that a majority portion was used for training the models, while the remaining portion was reserved for testing.

In this project, classification is the primary data mining technique used to categorize waste images into two predefined classes: recyclable and non-recyclable. Classification is a supervised learning technique where a model learns patterns from labeled data and predicts the class of unseen instances. Given the binary nature of the problem and the availability of labeled image data, classification was considered the most suitable technique for this task.

Three classical machine learning algorithms were selected: Logistic Regression, Decision Tree, and Random Forest. These algorithms were chosen because of their simplicity, interpretability, and effectiveness in small to medium datasets.

Logistic Regression works by modeling the probability of an instance belonging to a particular class using a logistic (sigmoid) function. It is well-suited for binary classification problems and provides probabilistic outputs, making it easy to interpret the influence of

input features. The sigmoid function can be give as:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

The pseudo-code for Logistic Regression can be given as:

```
Input: Feature matrix X, target labels y
Initialize model parameters (weights and bias)
For each iteration:
    Compute predicted probabilities using sigmoid function
    Calculate loss using log-loss function
    Update parameters using gradient descent
Output: Trained Logistic Regression model
Predict class labels for test data based on probability threshold
```

Decision Tree is a rule-based model that recursively splits the dataset based on feature values to maximize class purity. Its tree-like structure makes it highly interpretable. The pseudo-code explaining the algorithm can be given as:

```
Input: Feature matrix X, target labels y
Start with the full dataset as root node
If stopping criteria met:
    Assign class label
Else:
    Select best feature based on impurity measure
    Split dataset into child nodes
    Repeat process recursively for each child
Output: Decision Tree model
Predict class labels by traversing the tree
```

Random Forest is an ensemble method that combines multiple decision trees to improve generalization and reduce overfitting, making it more robust to noise and feature variability. The pseudo-code explaining the algorithm can be given as:

```
Input: Feature matrix X, target labels y
For each tree in the forest:
    Create a bootstrap sample from training data
    Train a Decision Tree using a random subset of features
Aggregate predictions from all trees
Output: Random Forest model
Final prediction obtained by majority voting
```

The input to all models consists of numerical features extracted from waste images, including color features (RGB means), texture features (LBP and GLCM), and shape features (HOG). The target variable is a binary label indicating whether the waste item is recyclable or non-recyclable. Key parameters include regularization strength for Logistic Regression, tree depth for Decision Trees, and the number of estimators for Random Forests. The expected output is a predicted class label along with performance metrics indicating classification accuracy.

6.5 Interpretation and Evaluation

To assess the performance of the classification models, several evaluation metrics were used. These metrics provide a comprehensive understanding of how well the models

classify recyclable and non-recyclable waste.
The primary evaluation metric used is Accuracy, defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where,

TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives

Precision measures the proportion of correctly predicted positive instances among all predicted positives.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall indicates the ability of the model to correctly identify all actual positive instances.

$$\text{Recall} = \frac{TP}{TP + FN}$$

The F1-score provides a balance between precision and recall, especially useful when both metrics are equally important. It is the Harmonic Mean of Precision and recall.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The confusion matrix was used to visualize the classification results, showing the number of correctly and incorrectly classified samples for each class. This helped in identifying whether the model was biased toward recyclable or non-recyclable waste.
The experimental results indicate that all three models achieved comparable accuracy, with the Decision Tree model showing slightly better balance between precision and recall. Logistic Regression performed consistently but was sensitive to outliers, while Random Forest provided stable performance due to its ensemble nature.

Key insights derived from the evaluation include the importance of effective feature extraction and preprocessing in image-based classification tasks. Although deep learning was not used, classical models demonstrated reasonable performance when combined with well-engineered features. The results suggest that such lightweight models can be effectively used in real-world waste classification systems where computational simplicity and interpretability are important.

7 Results and Insights

7.1 Results

The performance of Logistic Regression can be summarized by the following classification report

```
    Accuracy Score: 0.6414473684210527
                 precision    recall  f1-score   support

Non_Recyclable        0.64      0.68      0.66       157
    Recyclable        0.64      0.60      0.62       147
```

```

      accuracy                           0.64       304
     macro avg       0.64      0.64      0.64       304
  weighted avg       0.64      0.64      0.64       304
```

The following Confusion matrix details the performance of the model with the data.

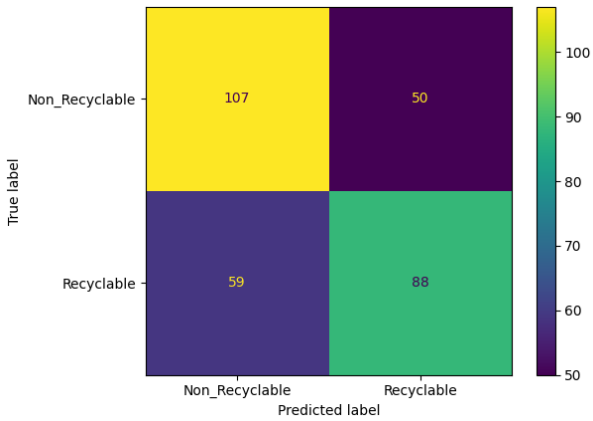

Figure 5: Confusion Matrix of Logistic Regression Model

The performance of Decision tree on the dataset can be summarized by the following classification report:

```
    Accuracy Score: 0.6480263157894737
                precision    recall  f1-score   support

Non_Recyclable       0.66      0.67      0.66       157
    Recyclable       0.64      0.63      0.63       147

      accuracy                           0.65       304
     macro avg       0.65      0.65      0.65       304
  weighted avg       0.65      0.65      0.65       304
```

The following confusion matrix details the performance of the model with the data.

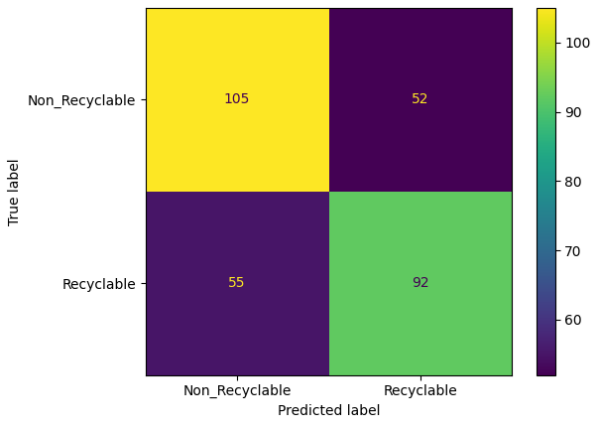

Figure 6: Confusion Matrix of Decision Tree Model

The performance of Random Forest on the dataset can be summarized by the following classification report:

```
    Accuracy Score: 0.6447368421052632
                  precision    recall  f1-score   support

Non_Recyclable       0.65      0.66      0.66       157
    Recyclable       0.63      0.63      0.63       147

      accuracy                           0.64       304
     macro avg       0.64      0.64      0.64       304
  weighted avg       0.64      0.64      0.64       304
```

The following confusion matrix details the performance of the model with the data.

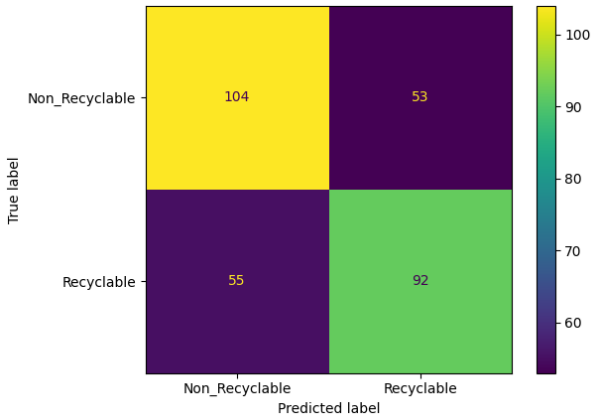

Figure 7: Confusion Matrix of Random Forest Model

7.2 Insights and discussions

The performance of the models was evaluated using accuracy, precision, recall, and F1-score. Logistic Regression achieved an accuracy of 64.14%. The model showed balanced precision for both recyclable and non-recyclable classes, but the recall for recyclable waste was slightly lower. This indicates that Logistic Regression is able to capture general patterns in the data but has limitations in handling complex variations present in image-based features.

The Decision Tree model performed slightly better with an accuracy of 64.80%, which was the highest among the three models. It showed improved balance between precision and recall for both classes. Since Decision Trees can model non-linear relationships, they were better able to separate recyclable and non-recyclable waste based on color and texture features extracted from the images.

Random Forest achieved an accuracy of 64.47%, which is comparable to the other models. The precision and recall values for both classes were well balanced, indicating stable performance. However, the improvement over Decision Tree was minimal. This suggests that increasing model complexity alone does not significantly improve performance when using handcrafted features.

Overall, all three models showed similar performance, with Decision Tree giving the best results for this dataset. The moderate accuracy values indicate that waste classification using manually extracted features is challenging, and better performance could be achieved using larger datasets or deep learning-based approaches.

8 Challenges, Pitfalls & Drawbacks

- **Data Collection Constraints:** Primary data collection was limited due to time and resource constraints, leading to reliance on secondary image datasets, which may affect real-world generalization.
- **Feature Overlap:** Significant overlap was observed between recyclable and non-recyclable classes in color and texture features, limiting perfect class separation.
- **Algorithm Limitations:** Classical machine learning models were used instead of deep learning approaches, which may restrict performance on large and diverse image datasets.
- **Image Variability:** Differences in background, lighting, and object orientation introduced noise and reduced feature consistency.
- **Limited Dataset Size:** Only 1,011 images were used for model training and evaluation, which restricts the ability of the model to learn diverse visual patterns and may hinder predictive performance.

9 Conclusion

Through this project, classical machine learning algorithms were successfully applied for the classification of household waste into recyclable and non-recyclable categories. The project leveraged feature extraction and engineering techniques and the Knowledge Discovery Database process to extract relevant features from the image database and preprocess it. Classical Machine Learning models were trained and evaluated, achieving interpretable results.

The project offers several opportunities for improvement and extension. increasing the size and complexity of the dataset for better prediction accuracy through additional primary data collection is the primary future scope of the project. Furthermore, advanced feature selection methods and deep learning approaches such as Convolutional Neural Networks (CNN) may further improve the classification accuracy. Additionally, the system can be extended into a real-world applications such as smart waste segregation systems, contributing to sustainable waste management practices and supporting environmentally responsible decision-making.

10 Discussion Questions

1. How does the choice of handcrafted feature extraction techniques such as HOG, LBP and GLCM influence the performance of Classical Machine

Learning models in image-based waste classification?
Machine learning models understand and process numerical data and not raw images Thus, handcrafted features play a crucial role in transforming the raw images into a structured numerical data. HOG captures shape and edge information, LBP represents local texture patterns, and GLCM extracts spatial texture relationships. Together, these models enhance the class separability. However, their effectiveness depends on the quality of images and may not capture the complex patterns as effectively as deep learning.

2. To what extent does dataset size and diversity affect the generalizability of waste classification models trained using traditional machine learning approaches?
Dataset size and quality crucially affects the generalizability of waste classification. If we have a dataset where there is a class imbalance, the model algorithm performance automatically declines and it will misclassify. A small or less diverse dataset limits the model's exposure to variations in lighting, background, and object orientation, leading to moderate performance on unseen data. . Increasing dataset size improves the model's ability to learn robust patterns and reduces overfitting.

3. Why do classical machine learning models show moderate performance despite careful preprocessing, and how could advanced techniques improve class separability?
Classical models rely heavily on manually extracted features, which may not fully represent complex visual patterns in waste images. Even with preprocessing, feature overlap between classes can reduce classification accuracy. Advanced techniques such as Convolutional Neural Networks can automatically learn hierarchical features, potentially improving class separability and predictive performance.

4. How can automated waste classification systems contribute to sustainable waste management practices in real-world scenarios?
Automated waste classification systems can improve waste segregation efficiency by reducing human effort and error. Accurate classification enables better recycling processes, minimizes landfill waste, and supports sustainable waste management initiatives. Such systems can be integrated into smart bins and recycling facilities to promote environmental sustainability.

5. What are the trade-offs between model interpretability and predictive performance when using classical machine learning algorithms instead of deep learning models?
Classical machine learning models are easier to interpret and computationally efficient, making them suitable for educational and resource-constrained environments. However, they may offer lower predictive performance compared to deep learning models, which achieve higher accuracy at the cost of increased complexity and reduced interpretability.

References & Bibliography

References

[1] S. B. Kotsiantis, "Supervised Machine Learning: A Review of Classification Techniques," *Informatica*, vol. 31, pp. 249–268, 2007.

[2] N. Dalal and B. Triggs, "Histograms of Oriented Gradients for Human Detection," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2005.

[3] T. Ojala, M. Pietikäinen, and D. Harwood, "A Comparative Study of Texture Measures with Classification Based on Featured Distributions," *Pattern Recognition*, vol. 29, no. 1, pp. 51–59, 1996.

[4] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural Features for Image Classification," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. SMC-3, no. 6, pp. 610–621, 1973.

[5] A. Rao, "Multi-Class Waste Classification Image Dataset for CNN Training and TFLite Deployment on Edge Devices," *IEEE Dataport*, June 30, 2025, doi:10.21227/mqyq-wd15.
https://ieee-dataport.org/documents/multi-class-waste-classification-image-dataset

[6] Y. Kanetkar, *Let Us Python*, 1st ed. New Delhi, India: BPB Publications, 2019.

A Project Metadata

- **Difficulty Level:** Intermediate
- **Domain:** Environmental Sustainability / Waste Management
- **Dataset:** Secondary (Image dataset) with limited primary samples
- **Tools:** Python
- **KDD Focus:** Preprocessing, Modelling, Analysis, and Interpretation
- **Author Contribution:**

Author Contributions

Aryan was primarily responsible for problem formulation, dataset preparation, feature extraction, model implementation, performance evaluation, and interpretation of results. Aryan also led the documentation, analysis, and overall integration of the project components.
Gaurav contributed to literature review, background study, and assisted in data organization and preliminary analysis. Gaurav also supported the documentation process and provided feedback during model comparison and result discussion.